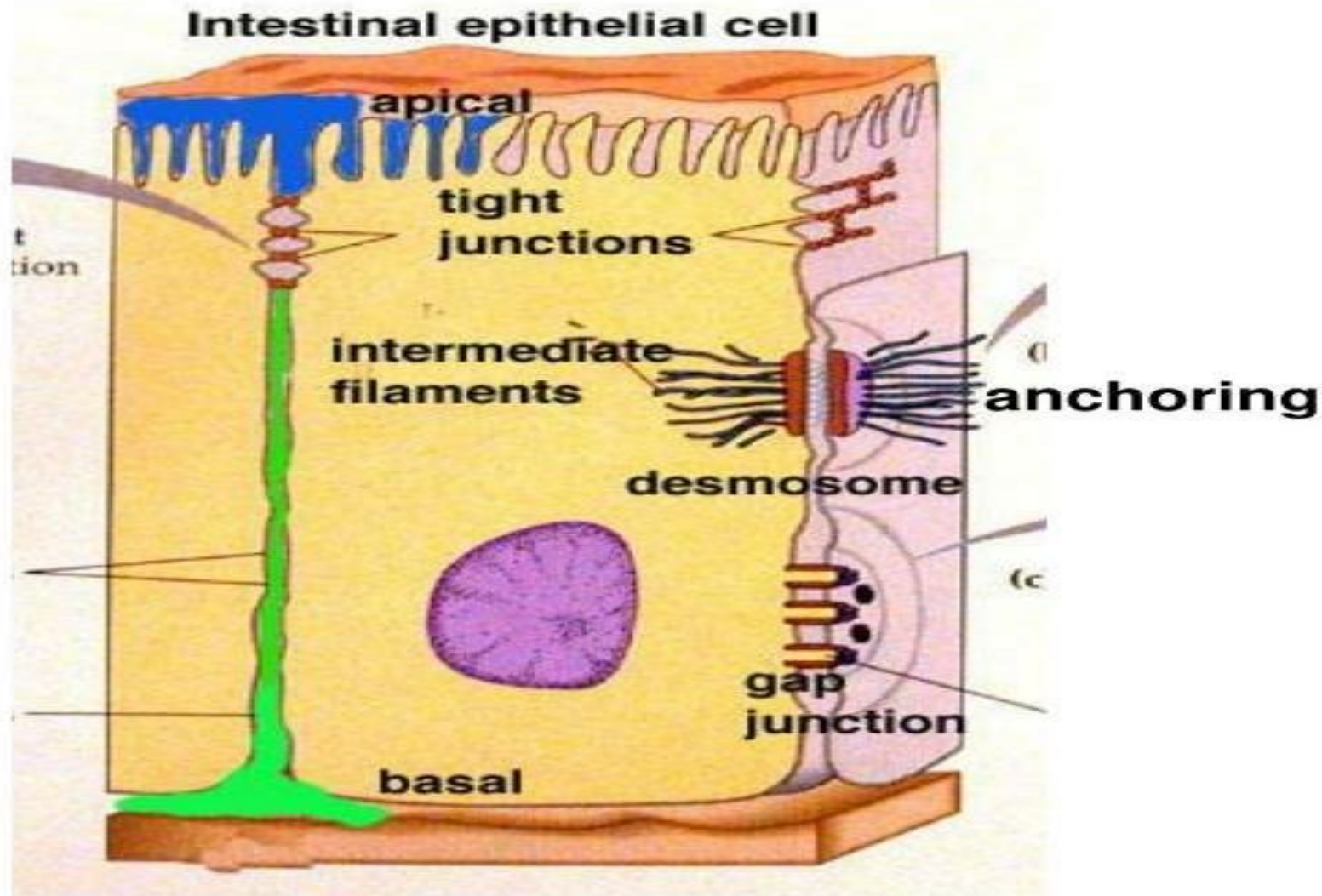


Cell Junctions



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Cell Junction

- **Cell junctions are contact points** between the plasma membranes of tissue cells.
- Five most important types of cell junction

1. Tight Junction

2. Adherens Junction

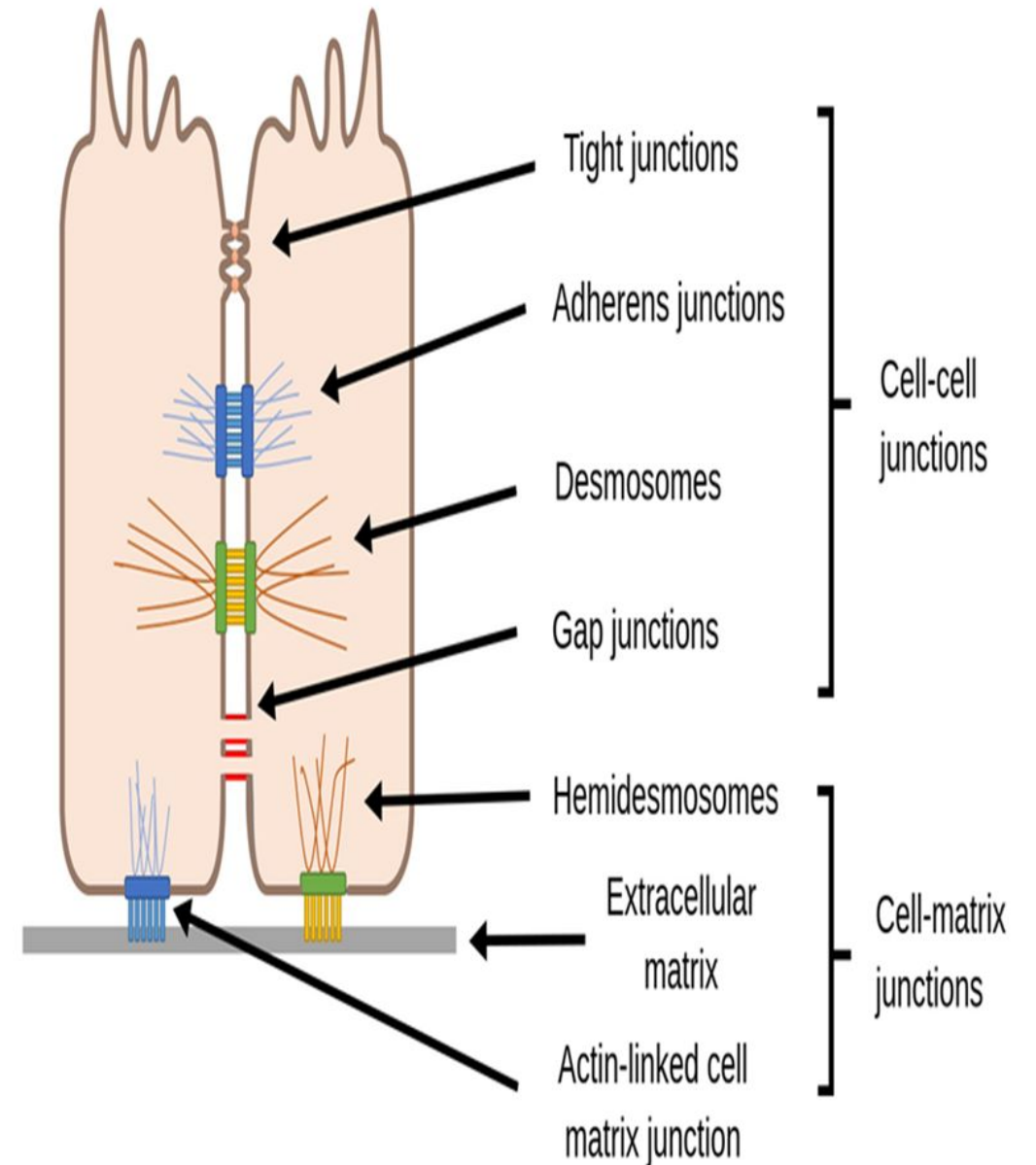
3. Desmosomes

4. Hemi desmosomes

5. Communication Junctions

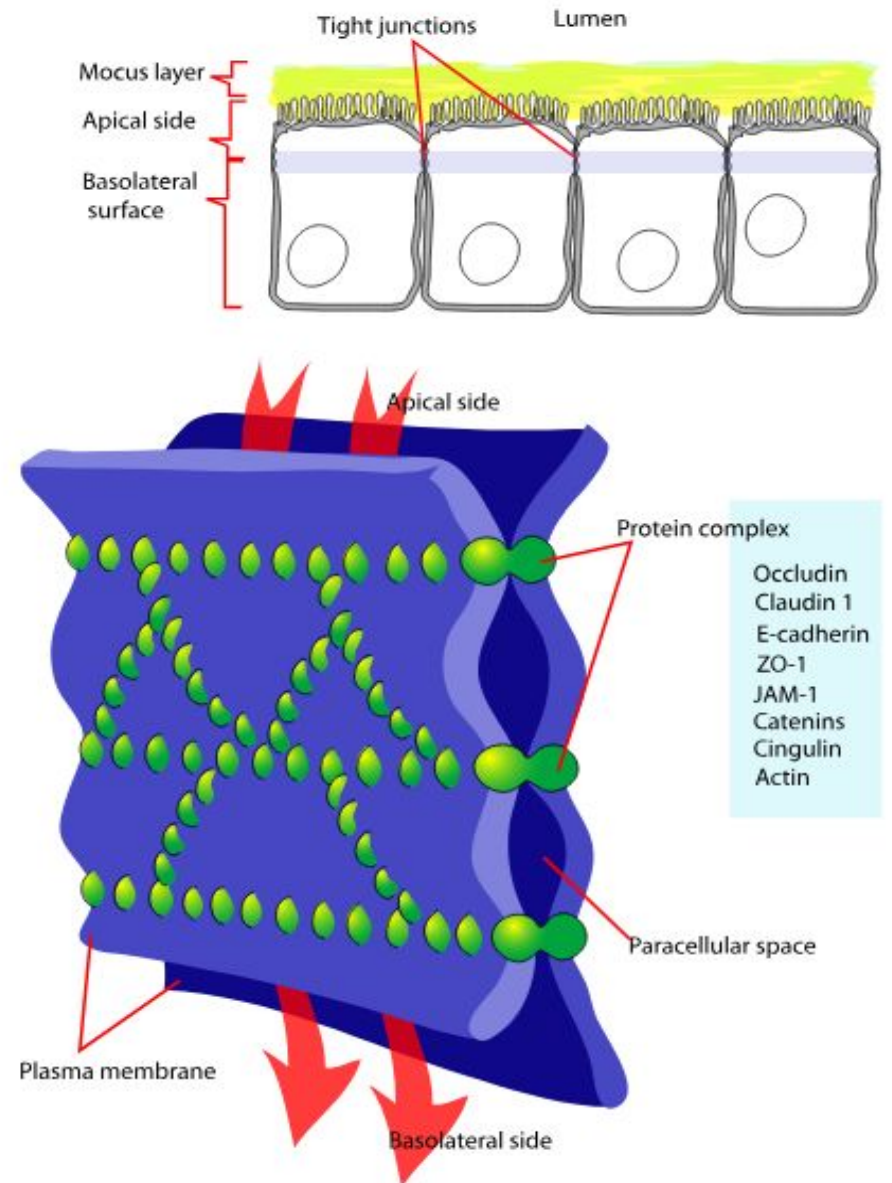
A- Gap Junction

B- CHEMICAL SYNAPSE



Tight Junction

Tight junctions (also called zonula occludens or occluding junctions) are a type of cell junction characterized by forming an adhesion complex between two neighboring cells serving as a tight seal between the cells.



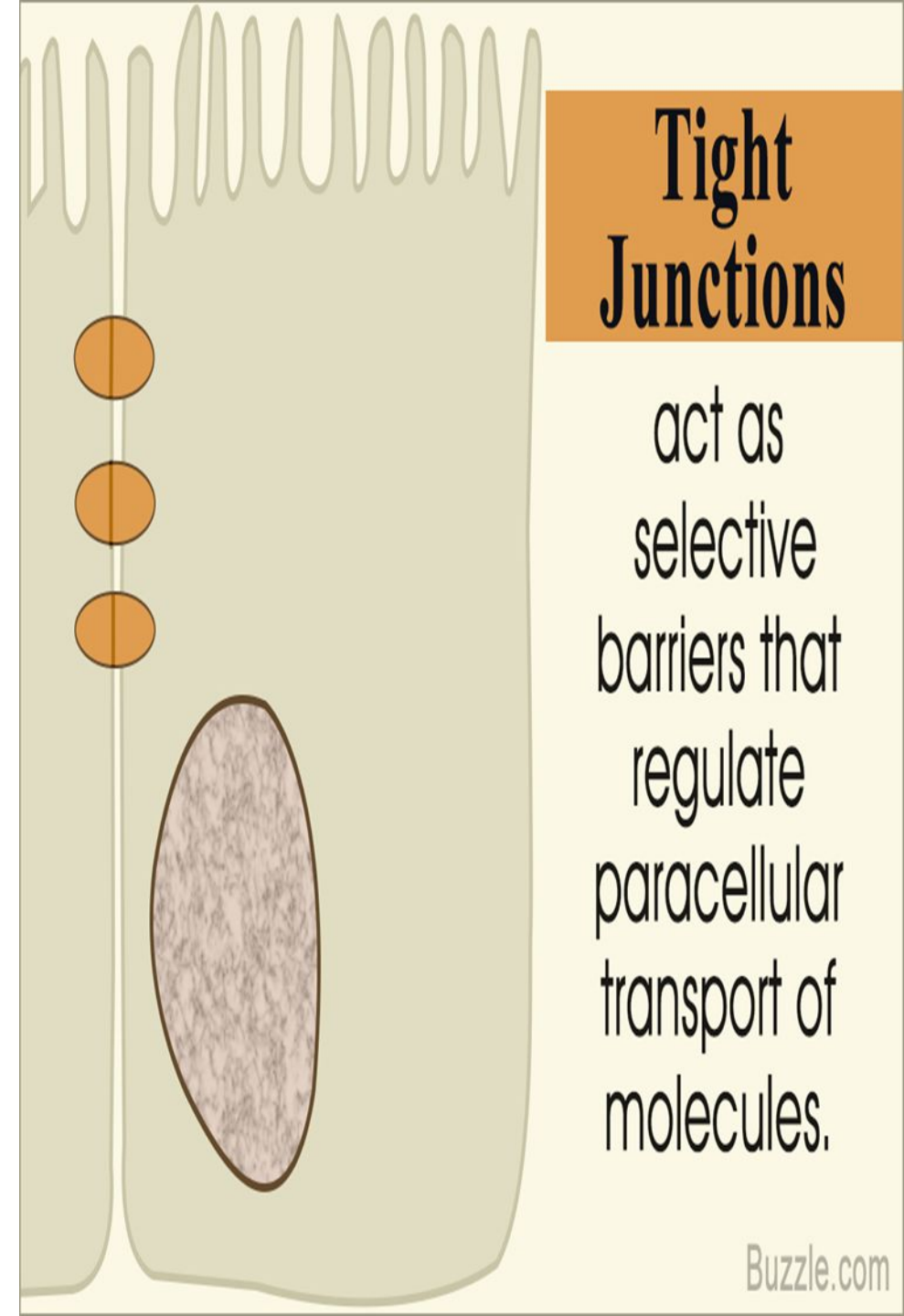
Structure of Tight Junction

Tight junctions consist of web like strands of transmembrane proteins that fuse together the outer surfaces of adjacent plasma membranes to seal off passage ways between adjacent cells.

It is also Cell Junctions called zonula occludes.

This type of junction is present in Cells of epithelial tissue that lines the stomach, choroid plexus intestines, and urinary bladder have many tight junctions.

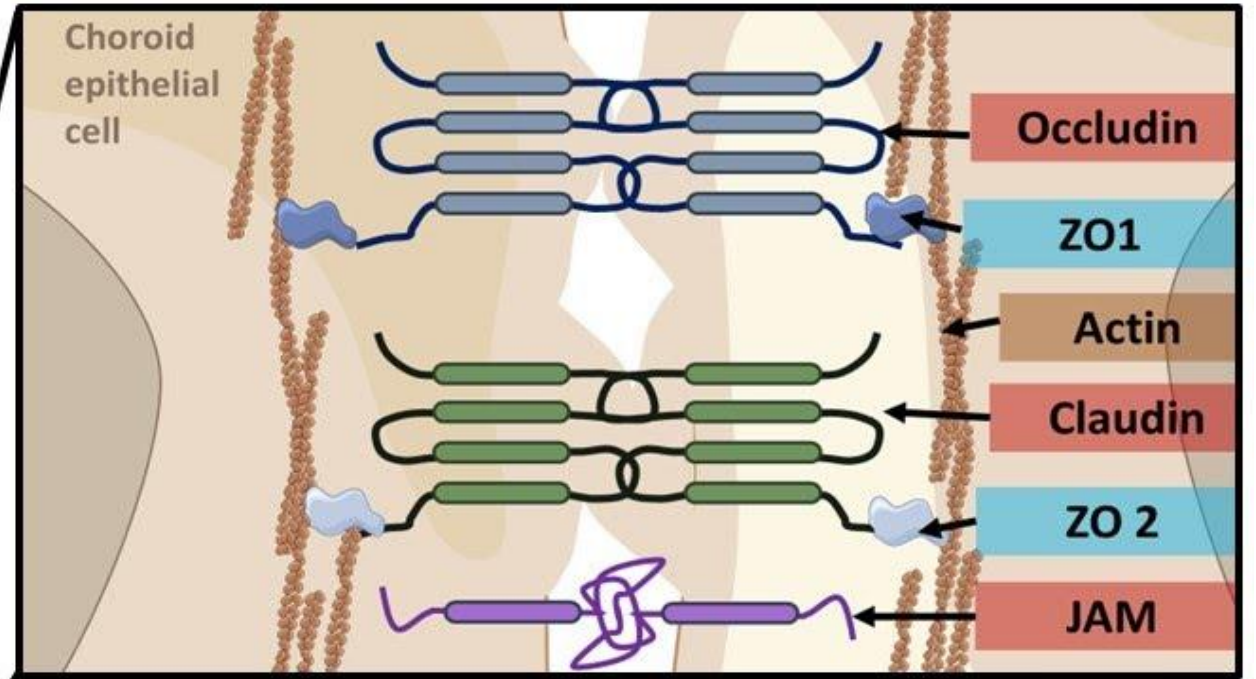
They inhibit the passage of substances between cells and prevent the contents of these organs from leaking into the blood or surrounding tissues.



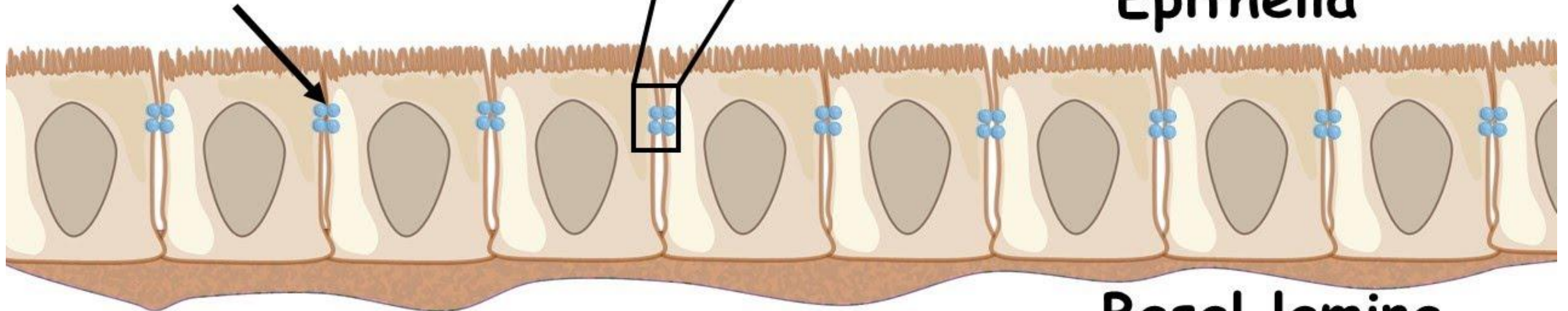
Tight Junctions

act as selective barriers that regulate paracellular transport of molecules.

Tight junction



Epithelia



Basal lamina

FUNCTIONS OF THE TIGHT JUNCTION

- **1- Strength and stability**

They hold the cells together

- **2- Barrier Function:-**

Tight junction in the brain capillaries forms the blood brain barrier, which prevents the entrance of many substances from capillary blood into brain tissues

- **3-Maintain polarity of the cell**

Tight junctions help to maintain the polarity of cells by preventing the lateral diffusion of integral membrane proteins between the apical and lateral/basal surfaces

- **4-Fencing function**

Tight junction prevent the passage of molecules and ions through the space between plasma membranes of adjacent cells

APPLIED PHYSIOLOGY

Diseases caused by mutation of genes encoding proteins of tight junction:

1. Hereditary deafness
2. Ichthyosis (scaly skin)
3. Sclerosing cholangitis (inflammation of bile duct causing obstruction)
4. Hereditary hypomagnesemia (low level of magnesium in the blood)
5. **Synovial sarcoma** (soft tissue cancer)

Functions of tight junction are affected by some bacteria and viruses also.

Proteins of tight junction

Proteins involved in the formation of tight junctions are classified into two types:

1. **Tight junction membrane proteins** or **integral membrane proteins**, such as occludin, claudin and junctional adhesion molecules (JAMs)
2. **Scaffold (framework or platform) proteins** or **peripheral membrane proteins** or cytoplasmic plaque proteins such as cingulin, symplekin and ZO1, 2, 3.

The **scaffold (platform) proteins** are attached with the **tight junction** membrane proteins and **strengthen** the anchoring in the ridges.

ADHERENS JUNCTION

Adherens junction is the cell to cell junction, which connects the actin filaments of one cell to those of another cell.

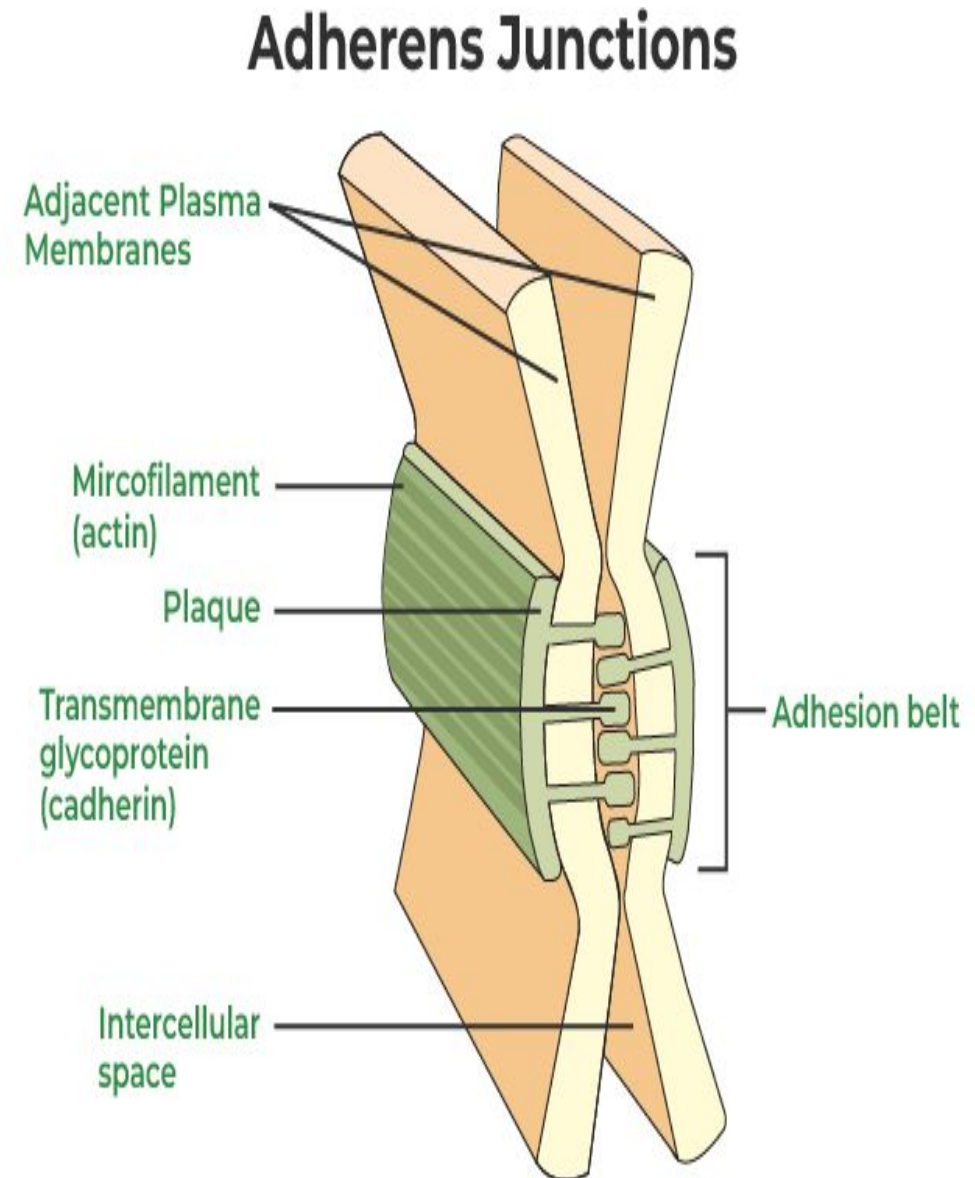
In adherens junction, the membranes of the adjacent cells are held together by some transmembrane proteins called **cadherins**.

Adherens junction provides strong mechanical attachments of the adjacent cells.

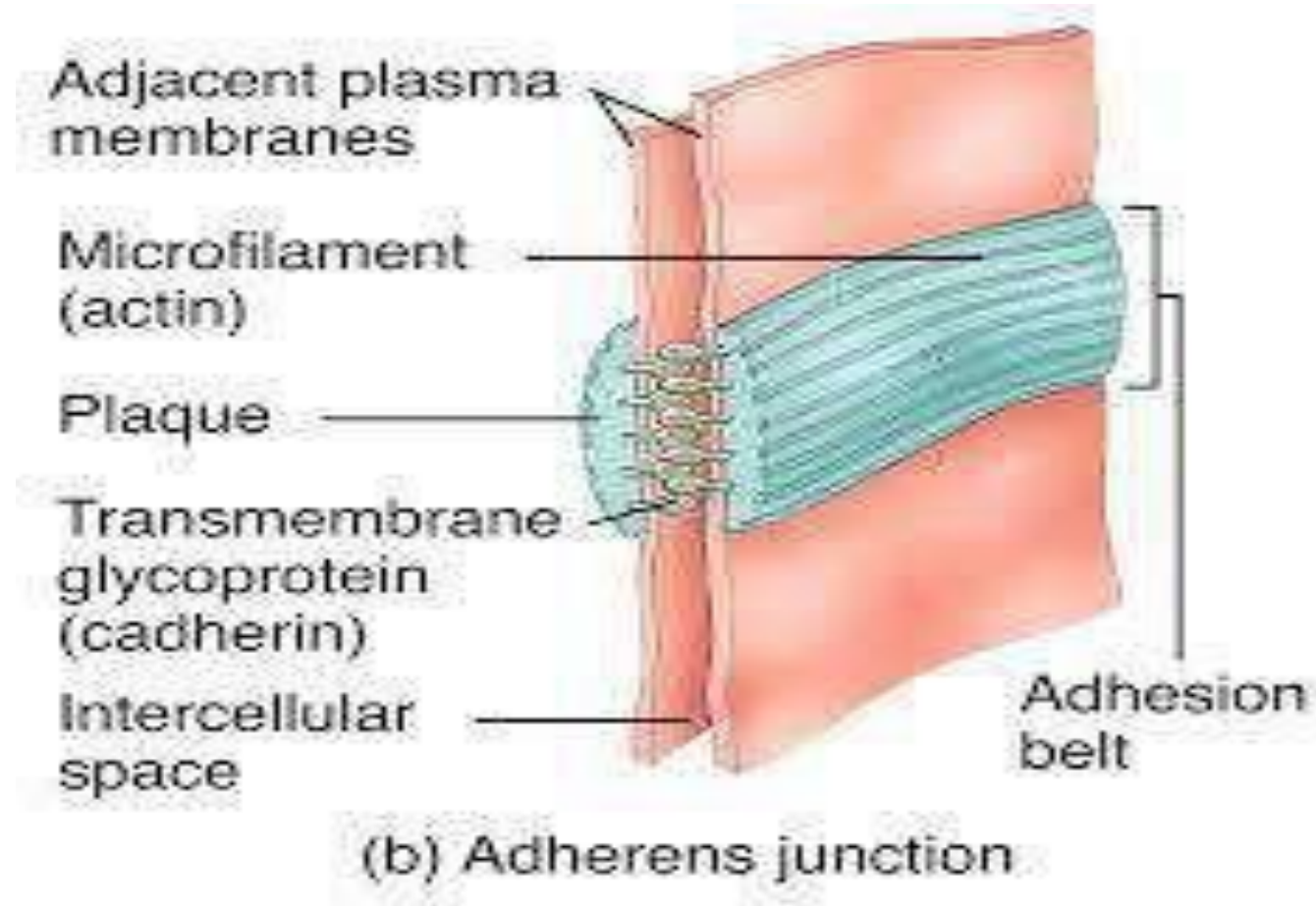
Adherens junction is present in the intercalated disks between the branches of cardiac muscles.

During the contractions and relaxation of heart, the cardiac muscle fibers are held together tightly by means of this junction.

The adherens junction present in epidermis helps the skin to withstand the mechanical stress.



Adherens Junction



Desmosomes

Desmosomes are a type of cell-to-cell junction, specialized intercellular structures that maintain tissue cohesion and structural integrity under mechanical stress.

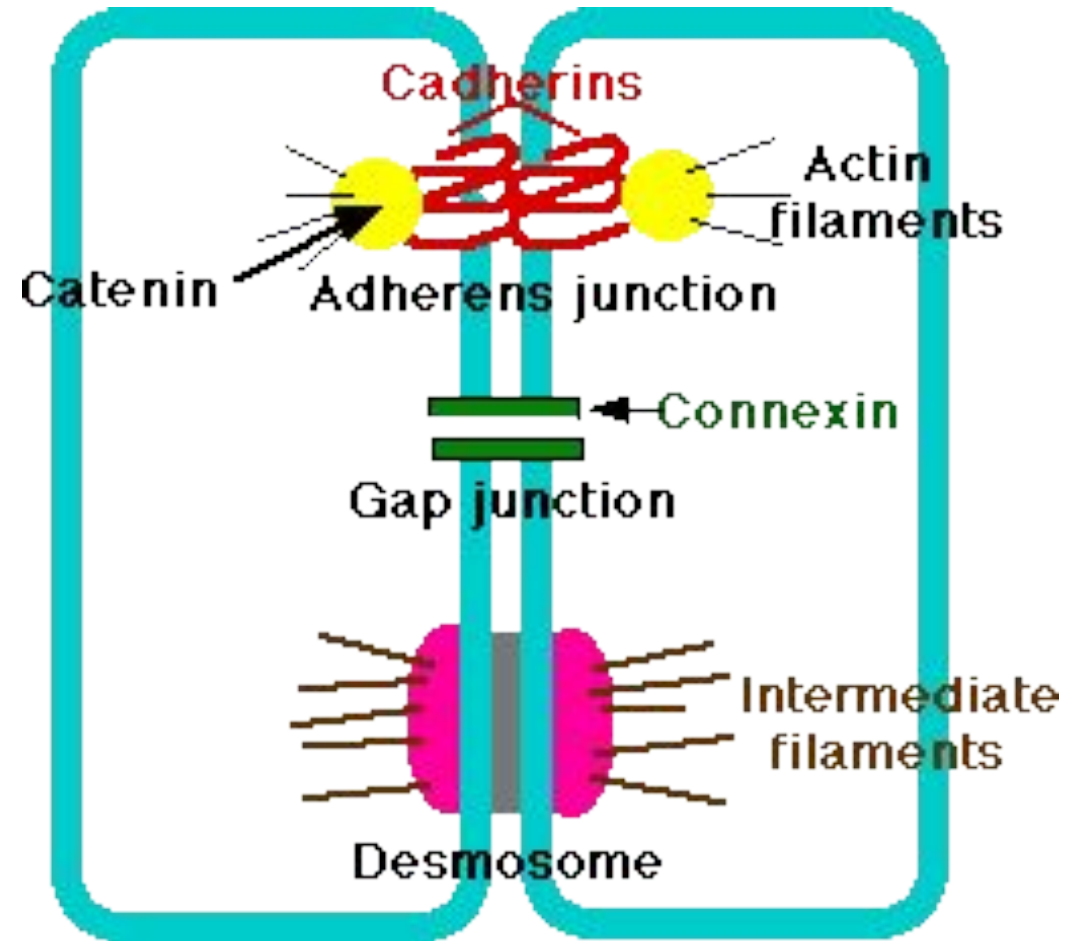
Desmosome is also called **macula adherens**.

Like adherens junctions, desmosomes contain plaque and have cadherins that extend into the intercellular space between adjacent cell membranes and attach cells to one another.

Desmosomes function like tight junctions.

The transmembrane proteins involved in desmosome are mainly cadherins.

Desmosomes



HEMIDESMOSOME

A hemi desmosome is a specialized junction that connects epithelial cells to the basement membrane, providing essential stability to the cells.

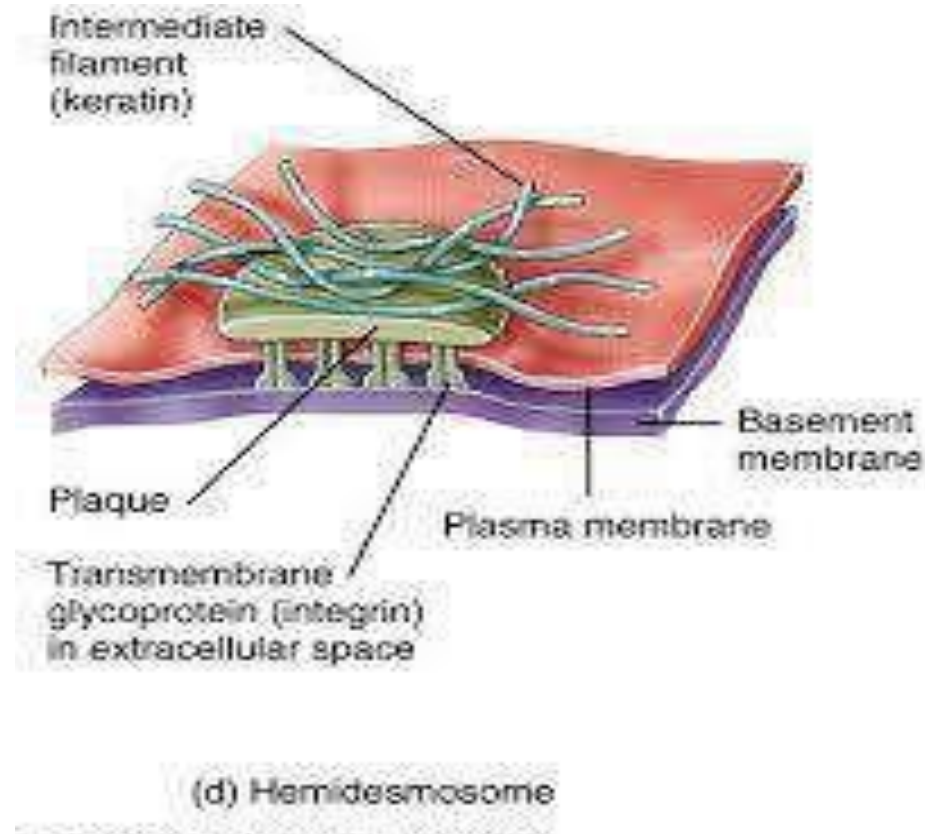
It is a **cell to matrix junction**, which connects the **intermediate filaments** of the cell to the **extracellular matrix**.

They provide essential structural stability, resisting mechanical shear and stress, while also facilitating cell polarization, migration, and signaling

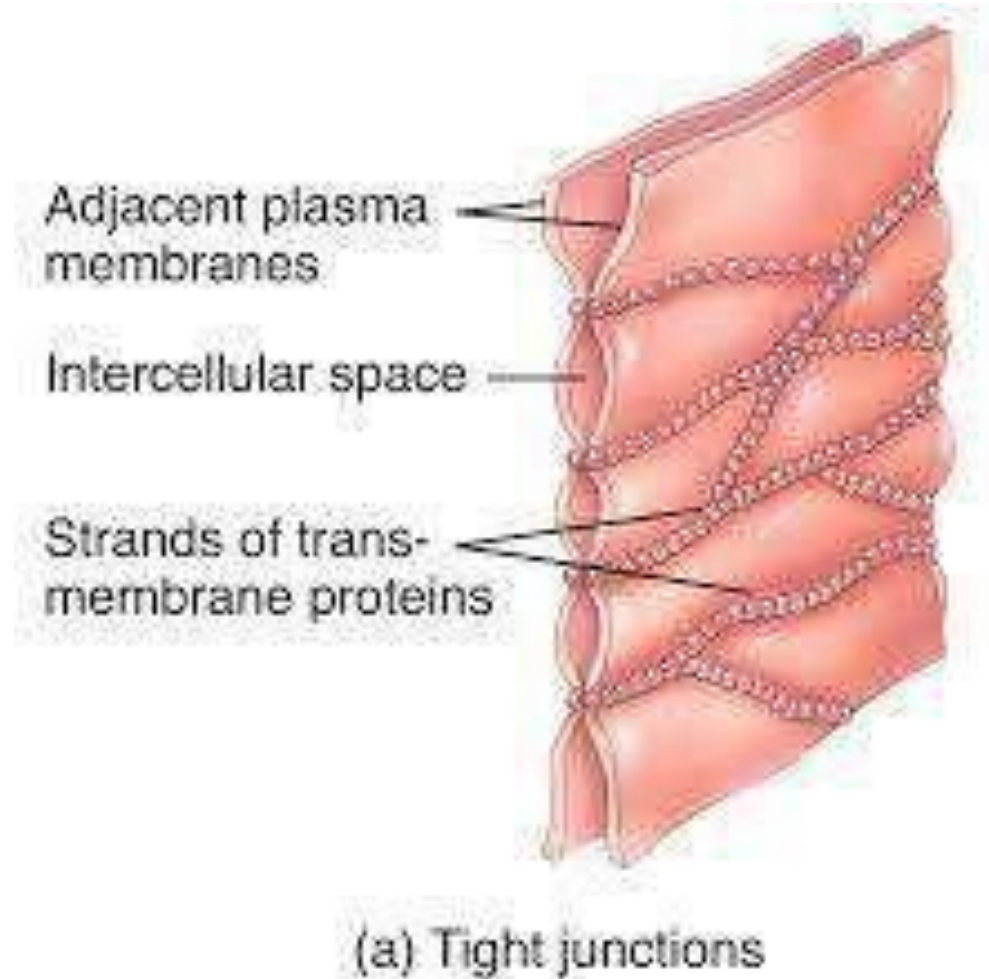
The proteins involved in this are **integrins**.

Desmosomes are specialized adhesive protein complexes that localize to intercellular junctions and are responsible for maintaining the mechanical integrity of tissues.

Hemidesmosomes



Tight Junction



COMMUNICATING JUNCTIONS

COMMUNICATING JUNCTIONS

Cell junctions which permit the intercellular exchange of substances are called communicating junctions, i.e. these junctions permit the movement of ions and molecules from one cell to another cell.

1-Gap junction and
2-chemical synapse are the communicating junctions.

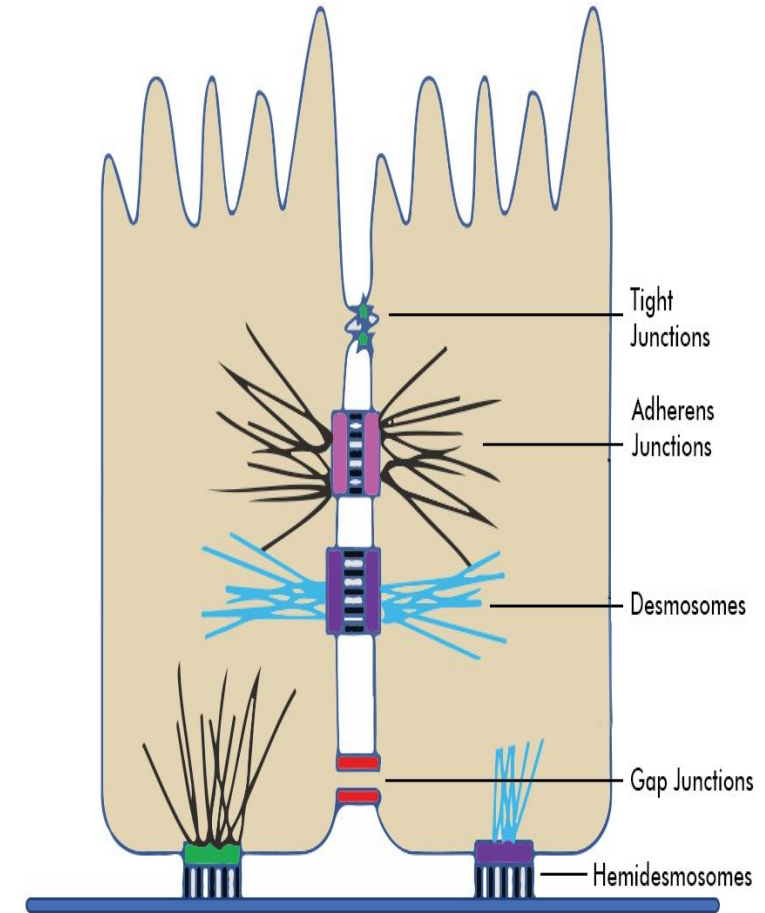
GAP JUNCTION:- Gap junction is the intercellular junction that allows passage of ions and smaller molecules between the cells. It is also called **nexus**.(Center Point)
It is present in **heart, basal part of epithelial cells of intestinal mucosa**, etc

Structure of Gap Junction

Membranes of the two adjacent cells lie very close to each other and the intercellular space is reduced from the usual size of 2.5 to 3 nm.

Cytoplasm of the two cells is connected by **the channels** formed by the membranes of both cells.

So, the molecules move from one cell to another cell directly through these channels, without having contact with extracellular fluid (ECF).



Functions of Gap Junction

- 1-Diameter of the channel in the gap junction is about **1.5 to 3 nm**.
So, the channel permits the passage of **glucose, amino acids, ions** and other substances, which have a molecular weight less than 1,000
- 2-It helps in the **exchange of chemical messengers** between the cells
- 3-It helps in rapid propagation of action potential from one cell to another cell
- 4-Gap junctions allow the exchange of ions, second messengers, and small metabolites between adjacent cells and are formed by two unrelated protein families, the **pannexins and connexins**.

Regulation of the Diameter of Channels in Gap Junction

In the gap junctions, the diameter of each channel is regulated by the intracellular **calcium ions**.

When the concentration of intracellular **calcium ion increases**, the **protein subunits of connexin** surrounding the channel come close to each other by **sliding**.

Thus, the **diameter of the channel decreases**.

The diameter of the channel is also regulated by **pH, electrical potential, hormones** or **neurotransmitter**.

CHEMICAL SYNAPSE

Chemical synapse is the junction between a nerve fiber and a muscle fiber or between two nerve fibers, through which the signals are transmitted by the release of chemical transmitter

APPLIED PHYSIOLOGY

Mutation in the genes encoding the connexins causes diseases such as:

1. **Deafness**
2. **Keratoderma** (thickening of skin on palms and soles)
3. **Cataract** (opacity of lens in eye)
4. **Peripheral neuropathy** (damage to the nerves of peripheral nervous system)
5. **Charcot-Marie Tooth disease** (a form of neuropathy)
6. **Heterotaxia** (abnormal arrangement of organs or parts of the body in relation to left right symmetry).

(Connexins constitute a large family of trans-membrane proteins that allow intercellular communication and the transfer of ions and small signaling molecules between cells).

APPLIED PHYSIOLOGY

1. Dysfunction of adherens junction and focal junction in colon due to mutation of proteins results in colon cancer.
 2. It also leads to tumor metastasis (spread of cancer cells from a primary tumor to other parts of the body)
 3. Dysfunction of hemidesmosome also causes bullous pemphigoid. (Bullous pemphigoid occurs when your immune system attacks a thin layer of tissue below your outer layer of skin. It is a autoimmune disease with tense blister eruptions of the skin) .
- The patients develop antibodies against cadherins.

CELL ADHESION MOLECULES

Cell adhesion molecules (CAMs) or cell adhesion proteins are the protein molecules, which are responsible for the attachment of cells to their neighbors or to basal lamina (or basal membrane).

CAMs form the important structures of intercellular connections and are responsible for structural organization of tissues.

TYPES OF CELL ADHESION MOLECULES

Cell adhesion molecules are classified into four types:

1. **Cadherins**, which form the molecular limbs between neighboring cells. These CAMs form adherens junction and desmosome
2. **Integrins**, which form the focal adhesion and hemidesmosome
3. **IgG super family**, which form the cell adhesion molecules in nervous system
4. **Selectins**, which act as receptors for carbohydrates (ligand or mucin) and are found in platelets and endothelial cell

Junction type	Protein Involved	Function	Example
Tight junction 3	Occludin Claudin JAMs Cingulin Symplekin ZO1, 2,	Strength and stability to tissues Selective permeability Fencing function Maintenance of cell polarity Formation of blood brain barrier	Epithelial lining of intestinal mucosa and renal tubule Endothelium in capillary wall and choroid plexus
Gap junction	Connexins	Allows passage of small molecules, ions and chemical messengers Propagation of action potential	Epithelial lining Heart Intestine
Adherens junction	Cadherins	Cell to cell attachment	Epithelial lining Heart Epidermis
Focal adhesions	Integrins	Cell attachment to Basal lamina Extracellular matrix	Epithelial lining
Desmosome	Cadherins	Cell to cell attachment	Epithelial lining Skin
Hemidesmosome	Integrins	Cell attachment to Basal lamina Extracellular matrix	Epithelial lining

THANK YOU